

22 – M2 Catalogue & Batch

Module design for the shared product truth: stock keeping units in one global namespace with local and shared status, units and pack conversions, barcodes many-to-one with optional batch resolution, tags, tax categories, product images, and physical batches carrying expiry and the printed maximum retail price. The module the data steward operates, and the largest reference set distributed to tills.

Field	Value
Document	22 – M2 Catalogue & Batch (Stage 2 module design; template of doc 20 §5)
Version	v0.1 – draft for design review; contract freeze targeted end of Stage 2 week 3
Date	8 September 2026
Owning developer	Module developer (full-stack): backend and web screens end to end; the till track consumes the catalogue snapshot; M4 and M5 command batch registration
Inputs	05 SRS (FR-SKU, FR-LOC-020); 16 answers (B section, P-07, R-02 to R-04, T-03); 09 ADR-07, ADR-08; 17 v0.2.1 §7.2; 18 v0.2 part B; 19 v0.2 §5, §8; 20 v0.3 M2 blueprint and Appendix A (images, attributes); 21 (owner entities, scopes); 30 §5.2; 32 §5.1; the September 2026 multi-MRP study (barcode-to-batch, 2D readiness)
Depends on	M1 (owner entity, scope, locations for assortment), kernel (audit, events, i18n, snapshot change log, attachments for images)
Depended on by	M3 (SKU, batch MRP, tags, tax), M4 (SKU and batch on documents; batch registration on GRN), M5 (batch identity, repack outputs), M6 (catalogue snapshot, barcode lookup), M7, M8, M10 (supplier table hand-over)

Why this module decides whether the programme lands. The requirements baseline names the assembly of a trusted catalogue as the critical path (T-03) and the data steward as the single largest open action. M2's screens are the steward's tools; if bulk import, duplicate detection across three scripts and translation completeness are afterthoughts, the catalogue will be assembled in spreadsheets and imported badly. They are first-class here.

1. Purpose and scope

M2 defines what can be counted and sold: the SKU as the smallest distinctly tracked item (B-01), its units and pack conversions, the barcodes that identify it at a scanner, the tags that group it, the tax category that prices it, the images that help a cashier recognise it, and the batch as the physical identity that carries expiry and printed MRP. It owns identity, not holding: where stock is and what it cost belong to M5 (*doc 17 change C4*).

In scope (phase 1)	Out of scope
SKU lifecycle in one global namespace: draft, local, shared, inactive; system-generated opaque codes; mandatory English short name, trilingual names for shared SKUs; bounded free attributes	Colour/size variant matrices, product families (each variant is a SKU by B-01)
Units of measure and per-SKU pack conversions (case of 24 is the same SKU); weighed goods by kilogram	Serialised items (B-05)
Barcode registry: many-to-one, symbology-aware, factory and in-store (20–29) codes, optional batch link; GS1 application-identifier data captured from 2D scans	Barcode printing hardware (M6 peripheral layer); label templates (M3 for price labels, M5 for batch stickers)
Tags as the governed grouping vocabulary; local tags per entity	A mandatory category hierarchy (B-15)
Tax categories and effective-dated rates (standard VAT)	Tax computation (shared engine) and invoicing (M4)
Product images and thumbnails; assortment definition per location for the till snapshot	Image recognition; images on receipts or customer displays
Batch registration from receipts and repacks; synthetic batches; MRP correction by replacement batch	Stock lots, movements, costs, expiry alerts (M5); price ceilings and control prices (M3)
Steward tooling: bulk import with validation, duplicate suspects across scripts, translation completeness, promotion with alias	Data cleansing itself (a named workstream, R-04)
A minimal supplier table until M10 takes it over	Supplier contracts, quotations, purchase orders (M10, phase 2)

2. Requirements coverage

Identifier	Requirement (abridged)	Design section
FR-SKU-010	SKU as smallest tracked item; global namespace; owner and status; opaque code; English name mandatory (B-01, B-08, B-09)	3.1, 4.1
FR-SKU-020	Case and single as one SKU with a conversion factor; a genuinely different item is a second SKU via repack (B-02, ADR-07)	3.2, 6.6
FR-SKU-030	Batch carries supplier, manufacture and expiry, printed MRP; tracking only where needed (B-04, ADR-08)	3.7, 4.3, 6.4
FR-SKU-040	Supplier attribution at batch level; one or two SKUs per product left to the owner (B-03)	3.7, 3.9
FR-SKU-050	Weighed goods by kilogram; bulk-to-pack via repack recipe (B-06)	3.2, 6.7
FR-SKU-060	Barcodes: not guaranteed; many-to-one; in-store 20–29 labels; code lookup (B-11, B-12)	3.3, 6.5
FR-SKU-070	Units and conversions defined per SKU by the owner (B-13)	3.2
FR-SKU-080	Trilingual master data with English fallback; data steward named (B-14, P-07, T-03)	3.1, 6.1, 6.2
FR-SKU-090	Tags instead of a mandatory hierarchy; governed vocabulary (B-15)	3.5
FR-SKU-100	Promote a local SKU to shared with an alias preserving history (B-10)	4.1, 6.3
FR-SKU-110	Tax category per SKU; effective-dated rates (G-09, J-03)	3.6
FR-LOC-020	Trilingual names in the catalogue; NFC and ICU collation	3.1, 9.4
B-17 (parked)	Deactivation while stock remains	4.1, 11 (DR-3)
Doc 20 Appendix A	Product images; bounded attributes jsonb	3.4, 3.1
Multi-MRP study	Barcode-to-batch link; 2D application-identifier capture	3.3, 6.5
R-02 to R-04	Opening catalogue from paper and spreadsheets; cleansing workstream; bulk import	6.1
NFR-LOC-010/020, NFR-PRF-040	Collation, search, snapshot size and delta	9

3. Domain model

Tables are those of doc 18 v0.2 part B owned by M2 (schema catalogue): uom, sku, sku_uom_conversion, sku_barcode, sku_alias, sku_image, tag, sku_tag, tax_category, tax_rate, batch; plus one table introduced here.

3.1 SKU

Aspect	Design
Aggregate	sku with its conversions, barcodes, images, tags and attributes. Root of identity for everything sold or counted.
Invariants	sku_code system-generated, opaque, unique, never reused. short_name_en non-null in every status; _si and _ta non-null for SHARED (B-I1). Owner is the Federation for SHARED; the creating entity for DRAFT and LOCAL. base_uom_code is KG when sold_by_weight. attributes under 4 KB, no behaviour, not searchable.
Visibility	SHARED SKUs are readable by every scope (the reference exception to OWN: a policy shared_read on status = SHARED). LOCAL and DRAFT SKUs are OWN only, and Federation view-only (M-03).
Flags that drive other modules	batch_tracked, expiry_tracked, has_printed_mrp (new; false for loose produce and unpriced goods so a null batch MRP is legitimate rather than missing), sold_by_weight, multi_mrp_policy (default from M3's policy; stored here because it is a SKU property).

3.2 Units and conversions

Aspect	Design
Vocabulary	uom is Federation-governed: EA, KG, G, L, ML, CASE, PKT, BAG, BOX, DOZ. An entity cannot add units; it can define conversions for its own SKUs.
Conversion	sku_uom_conversion(sku, uom, factor_to_base, effective_from/to): CASE = 24 × EA. A supplier's case-size change is a new effective-dated row, not an edit (P5). Weighed goods: base KG; retail packs of a bulk SKU are a different SKU produced by an M5 recipe, never a conversion (ADR-07).

3.3 Barcodes

Aspect	Design
Registry	sku_barcode(barcode, symbology, sku, uom, batch_id?, owner_entity, status). Unique per (barcode, symbology) across the federation for factory symbologies; INTERNAL codes unique within the owner entity (B-I2).
In-store codes	GS1 prefixes 20–29 for weigh labels and stickers. The label encodes SKU code and weight (or SKU code and batch), never a price, so the till resolves the price through the shared engine and the control-price ceiling still applies (<i>multi-MRP study</i> ; doc 18 §4.9).
Batch link	batch_id set when the code identifies one batch: an MPCS batch sticker (option C of the study), a manufacturer code that changed with the MRP (option B), or a 2D code carrying application identifier 10. The till's scan layer passes the parsed GTIN, lot and expiry; M2's lookup returns the SKU and, when it can, the batch.
2D readiness	LookupByBarcode accepts (gtin, lot?, expiry?, symbology); when lot is present and a batch with that number exists for the SKU, the batch is returned even without a registered batch barcode – this is what makes GS1 2027 marking pay off without any catalogue work.

3.4 Images

Aspect	Design
Model	sku_image(sku, barcode?, owner, object_key_full, object_key_thumb, hash, status) (doc 18 v0.2). One ACTIVE image per (sku, barcode-or-null).
Thumbnails	An M2 job (ImageThumbnailJob) produces the 128 px WebP thumbnail on attach; the snapshot carries only thumbnail keys; the till fetches lazily and caches (doc 20 M2; doc 32 §5.1).
Ownership	Federation attaches for SHARED SKUs; an entity may attach for its LOCAL SKUs; an entity may also attach a <i>local override</i> for a SHARED SKU (different pack in that region), scoped to its own locations.

3.5 Tags

Aspect	Design
Vocabulary	tag(tag_code, names, governed, owner_entity). Governed tags are Federation-owned and used for federation-wide comparison and for policy attachment (expiry windows, loss tolerances, variance thresholds, control-price scope, multi-MRP policy). Local tags belong to an entity for its own merchandising.

Aspect	Design
Rule	A policy in M3 or M5 may attach to a governed tag only; local tags carry no behaviour. Enforced by the consuming modules' validators through <code>TagQueries.isGoverned</code> .

3.6 Tax

Aspect	Design
Model	<code>tax_category(code, names)</code> and <code>tax_rate(category, rate_percent, effective_from/to)</code> . Standard VAT only (J-03); exempt and zero-rated as categories with a 0% rate row. Publication of a new rate is a Federation command with an <code>apply_from</code> that reaches tills through the snapshot (<i>doc 32 §5.2</i>).

3.7 Batch

Aspect	Design
Identity	<code>batch(sku, batch_no, supplier, manufacture_date, expiry_date, printed_mrp, origin_document, is_synthetic)</code> . Global, not owned: one physical batch is held successively by the Federation, a distributor and an MPCS; each holding is an M5 stock lot with its own cost (<i>doc 17 change C4, ADR-38</i>).
Uniqueness	<code>(sku_id, supplier_id, batch_no)</code> unique. A receiver who keys a batch that already exists gets the existing row (the GRN cites it); a receiver who keys a new number for the same physical batch creates a duplicate identity – accepted, flagged as a duplicate suspect for the steward (Section 6.2).
Synthetic batches	For SKUs with <code>batch_tracked = false</code> each receipt line creates a synthetic batch (<code>is_synthetic = true, batch_no = 'S-' + document number + line</code>) so that MRP and cost always have a home and price resolution has one code path (<i>doc 18 DR-5</i>).
MRP	<code>printed_mrp</code> mandatory when the SKU has <code>has_printed_mrp</code> ; immutable. A mis-keyed MRP is corrected by <code>CorrectBatchMrp</code> : a new batch row, an audited link <code>corrects_batch_id</code> , and M5 re-points the lots (B-I8). Two concurrent batches may legally carry different MRPs (ADR-08).
Who creates batches	M4's GRN confirmation (supplier goods at a Federation warehouse; distributor and MPCS receipts cite the incoming batch or create a local-supply batch) and M5's repack execution (output batch for the produced SKU). M2 exposes <code>RegisterBatch</code> as an internal command those modules call in their own transaction.

3.8 Assortment (new concept, for the snapshot)

Aspect	Design
Definition	The set of SKUs a location's till must know: SKUs on the owning MPCS's retail price list (M3 query) $\cup$ SKUs with a lot at the location (M5 query) $\cup$ LOCAL SKUs of the owning entity $\cup$ SKUs tagged <code>core-range</code> (a governed tag the Federation uses for goods every shop should be able to sell).
Materialisation	<code>location_assortment(location_id, sku_id, reason, since)</code> maintained by an M2 consumer of price-list, lot and SKU events; the snapshot contributor reads it; a change to it writes the change-log row for that location (<i>doc 32 §5.1</i>). Change request CR-22-2 adds the table to doc 18 part B.

3.9 Minimal supplier table (until M10)

Aspect	Design
Model	<code>supplier(supplier_id, owner_entity, name, status)</code> – enough for batch attribution (B-03) and Federation receipts. Change request CR-22-1 adds it to doc 18 with the note that M10 absorbs it in phase 2 (<i>doc 20 M10 seams</i>).

4. State machines

4.1 SKU

From	To	Trigger / command	Guard	Permission	Audit
–	DRAFT	CreateSku	English name; base unit; tax category; owner = caller scope	cat.sku.create_local (entity) or cat.sku.create (F)	SKU_CREATED
DRAFT	LOCAL	ActivateLocalSku	Owner entity; at least one barcode or code-lookup flag; images optional	cat.sku.create_local	SKU_ACTIVATED
DRAFT	SHARED	ActivateSharedSku	Caller F; three names; governed tags; conversion for CASE if applicable	cat.sku.create (F)	SKU_SHARED
LOCAL	SHARED	PromoteToShared (as new shared)	Caller F; three names complete; owner becomes Federation; originating entity keeps history via alias	cat.sku.promote (F)	SKU_PROMOTED
LOCAL	INACTIVE (aliased)	MergeIntoShared(target)	Caller F; target SHARED; alias row created; barcodes moved; the local SKU's lots re-point (M5)	cat.sku.promote (F)	SKU_MERGED
LOCAL / SHARED	INACTIVE	DeactivateSku	Reason; see DR-3 for stock-remaining behaviour	cat.sku.deactivate (owner)	SKU_DEACTIVATED
INACTIVE	LOCAL / SHARED	ReactivateSku	Reason	cat.sku.deactivate (owner)	SKU_REACTIVATED

Proposed rule for DR-3 (B-17): INACTIVE blocks new orders, purchases and price-list additions, but tills continue to sell lots on hand (the snapshot keeps INACTIVE SKUs with a `sell_through` flag while any lot exists at the location). This removes the classic failure of a till refusing to sell saleable goods.

4.2 Barcode

From	To	Trigger / command	Guard	Permission	Audit
–	ACTIVE	RegisterBarcode	Unique per symbology (federation) or per owner (INTERNAL); SKU not INACTIVE	cat.barcode.manage (owner of SKU; any entity for INTERNAL on its own SKUs)	BARCODE_REGISTERED
ACTIVE	RETIRED	RetireBarcode	Reason	cat.barcode.manage	BARCODE_RETIRED
ACTIVE	ACTIVE (relinked)	LinkBarcodeToBatch	Batch belongs to the SKU	cat.barcode.manage	BARCODE_LINKED

4.3 Batch

From	To	Trigger / command	Guard	Permission	Audit
–	REGISTERED	RegisterBatch (from M4 GRN or M5 repack)	SKU active; unique (sku, supplier, batch_no); MRP present when required; expiry when expiry-tracked	internal (caller module's permission)	BATCH_REGISTERED
REGISTERED	SUPERSEDED	CorrectBatchMrp / CorrectBatchExpiry	Reason; caller owns a lot of the batch or is F; creates the replacement batch	cat.batch.correct (owner) with MFA	BATCH_CORRECTED

4.4 Tax rate

From	To	Trigger / command	Guard	Permission	Audit
–	SCHEDULED	PublishTaxRate	Caller F; category exists; apply_from ≥ today; no overlap	cat.tax.publish (F, MFA)	TAX_RATE_PUBLISHED
SCHEDULED	IN_FORCE	Clock (apply_from)	–	system	–
IN_FORCE	SUPERSEDED	PublishTaxRate (next)	Sets effective_to on the current	cat.tax.publish	TAX_RATE_PUBLISHED

5. Contract

5.1 Commands

Command	Input (essentials)	Permission	Effect	Events
CreateSku	names, base uom, tax category, flags, tags, attributes	cat.sku.create_local / cat.sku.create (F)	DRAFT with generated code	sku.created.v1
ActivateLocalSku / ActivateSharedSku	sku	as above	LOCAL / SHARED	sku.activated.v1 / sku.shared.v1
UpdateSku	names, descriptions, flags, attributes, tags	owner; F for SHARED	Fields (audited diff); changing base uom or tracking flags requires no lots anywhere (M5 query)	sku.updated.v1
PromoteToShared	local sku, completed names, governed tags	cat.sku.promote (F)	Owner → Federation; alias row for the origin entity	sku.promoted.v1
MergeIntoShared	local sku, target shared sku	cat.sku.promote (F)	Alias; barcodes moved; local INACTIVE	sku.merged.v1
DeactivateSku / ReactivateSku	sku, reason	cat.sku.deactivate	Status	sku.deactivated/re-activated.v1
DefineConversion	sku, uom, factor, effective_from	owner	New effective-dated row	conversion.defined.v1
RegisterBarcode / RetireBarcode / LinkBarcodeToBatch	barcode, symbology, sku, uom, batch?	cat.barcode.manage	Registry row	barcode.registered/retired/linked.v1
AttachImage / RetireImage	sku, barcode?, object key, hash	cat.image.manage (owner)	Image row; thumbnail job	image.attached/retired.v1
DefineTag / TagSku / UntagSku	tag, names, governed (F only)	cat.tag.manage (F for governed; owner for local)	Tag rows	tag.defined.v1; sku.updated.v1
DefineTaxCategory / PublishTaxRate	category; rate, apply_from	cat.tax.publish (F, MFA)	Effective-dated rate	tax_rate.published.v1
RegisterBatch (internal)	sku, supplier, batch_no, dates, printed_mrp, origin document	caller module (M4, M5)	Batch row or existing row returned	batch.registered.v1
CorrectBatchMrp / CorrectBatchExpiry	batch, new value, reason	cat.batch.correct (MFA)	New batch with corrects_batch_id; old SUPERSEDED	batch.corrected.v1
RegisterSupplier	name, owner	cat.supplier.manage	Minimal supplier row	supplier.registered.v1
BulkImportSkus	file (sheet), mode (validate commit)	cat.sku.create / create_local	Validation report; DRAFT rows on commit	sku.created.v1 per row
ResolveDuplicateSuspect	suspect pair, decision (merge distinct)	cat.sku.promote (F) or owner for local pairs	Merge path or suppression record	sku.merged.v1 or duplicate.dismissed.v1

5.2 Queries

Query	Returns	Scope rule
LookupByBarcode(barcode gtin+lot+expiry, symbology, location)	sku, uom, factor, optional batch, thumbnail key	Any authenticated scope for SHARED; OWN for LOCAL; the till uses the snapshot copy offline
SearchSku(text, language, scope, status[])	Paged records ordered by the display language's collation; trigram match on all three names	As above
GetSku / ListSkus(filter)	Full record with conversions, barcodes, tags, images	As above; attributes OWN only
GetAssortment(location)	SKU ids with reasons	OWN, FEDERATION_VIEW; snapshot builder
GetCatalogueSnapshot(location, since_version)	Delta or full for the assortment	Snapshot builder only
ResolveAlias(sku)	Canonical sku	Any

Query	Returns	Scope rule
ListBatches(sku, filter)	Batch identities with expiry and MRP	OWN via lots (M5 join) or FEDERATION_VIEW
GetBatch(batch)	Identity record	As above
TagQueries.isGoverned(tag) / ListTags(scope)	Tags	Any
TaxQueries.rateInForce(category, date)	Rate	Any; also in the shared engine's snapshot
StewardDashboard()	Translation completeness, duplicate suspects, orphan barcodes, SKUs without images by tag	FEDERATION

5.3 Events published

Event	Payload (beyond envelope)	Consumers
sku.created/activated/shared/promoted/merged/updated/deactivated/reactivated.v1	sku id, code, owner, status, flags, tags, alias (for merged)	Assortment maintainer (M2), snapshot change log, M3 (policy defaults), M5 (alias re-pointing), M8
conversion.defined.v1	sku, uom, factor, effective_from	Snapshot, M4, M5
barcode.registered/retired/linked.v1	barcode, symbology, sku, uom, batch	Snapshot, M6 lookup
image.attached/retired.v1	sku, barcode, thumb key	Snapshot
tag.defined.v1	tag, governed	M3, M5 validators, M8
tax_rate.published.v1	category, rate, apply_from	Snapshot (apply_from), M3, M4, shared engine
batch.registered/corrected.v1	batch id, sku, supplier, batch_no, expiry, printed_mrp, corrects	M3 (MRP review vs list), M5 (lot re-point on correction), M8
supplier.registered.v1	supplier id, owner	M4, M10
duplicate.suspected/dismissed.v1	pair, score, reason	Steward dashboard, M8

6. Key flows

Flow	Sequence	Failure paths
6.1 Assembling the initial catalogue (the T-03 critical path)	Steward uploads the consolidated sheet in validate mode → report: rows accepted, rows with errors (missing English name, unknown unit, bad tax category), duplicate suspects across scripts (Section 6.2) → steward fixes the sheet or resolves suspects → commit creates DRAFT SKUs → steward completes translations (dashboard shows completeness by tag) → ActivateSharedSku in bulk for rows with three names → barcodes registered from the sheet → images attached in bulk by SKU code	Nothing partial: each row commits or is reported; a re-upload of the same sheet is idempotent by a row key (sheet row hash); suspects are never auto-merged
6.2 Duplicate detection across scripts	On import and on CreateSku: normalise names (NFC, case, punctuation), transliterate Sinhala and Tamil names to a Latin key with a fixed table, compare (brand tokens, pack size tokens, unit) and barcodes; score; pairs above a threshold become duplicate_suspect rows for the steward	A dismissed pair is remembered; thresholds are configuration; false positives cost a click, false negatives cost a permanent split in sales history
6.3 Promoting a local SKU	An MPCS's local item sells well; Federation adds it to the shared range → PromoteToShared (as new) or MergeIntoShared (existing shared SKU) → alias row (origin sku → canonical) → sku.promoted/merged.v1 → M5 re-points the origin entity's lots; M8 resolves aliases at query time so trends are unbroken (FR-SKU-100)	Merge with lots in a different base unit is refused; barcodes that collide with the target's are retired with a report
6.4 Batch registration at receipt	M4 confirms a GRN → per line: RegisterBatch(sku, supplier, batch_no, dates, MRP) → existing batch returned or created (synthetic when not batch-tracked) → batch.registered.v1 → M3 compares MRP with the standing list and raises a pricing review when lower	MRP missing for a has_printed_mrp SKU blocks GRN confirmation for that line with a clear message; a new batch_no for an existing physical batch creates a suspect, not an error
6.5 Scan resolution at the till (M2's part)	Scan layer parses (symbology, gtin, lot?, expiry?) → snapshot lookup: barcode → sku, uom, batch link; or gtin + lot → batch by number → returns item card data; price resolution is M3's engine	Unknown code → code-lookup or search by name; a code mapped to an INACTIVE SKU without sell-through → refuse with reason
6.6 Case and single	Supplier delivers cases; GRN line in CASE; conversion 24 → base EA; the till sells EA; a case barcode maps to (sku, CASE)	A missing conversion blocks the GRN line in that unit
6.7 Bulk to pack	50 kg bulk SKU (KG) is repacked into 1 kg pack SKU (EA) by an M5 recipe; the output SKU is a distinct SKU created here (origin REPACK_OUTPUT); the output batch is registered by M5 with the input batch's expiry and a local MRP if labelled	No conversion may link the two SKUs (validator)
6.8 MRP correction	Warehouse keyed 980 instead of 1,080 → CorrectBatchMrp with reason and MFA → new batch (corrects old) → batch.corrected.v1 → M5 re-points lots; M3 re-evaluates ceilings; receipts already issued keep their recorded mrp_applied	A correction after sales were capped by the wrong MRP is reported to the entity's exception queue (money was left on the table, or, if the MRP was too high, an over-charge review)

7. Configuration

Item	Default	Scope	Who may change
catalogue.duplicate_threshold	0.85	Federation	cat.sku.promote
catalogue.transliteration_table_version	v1	Federation	–
catalogue.thumbnail_px / max_thumb_kb	128 / 10	Federation	–
catalogue.local_sku_requires_barcode	false	Entity	cat.sku.create_local
catalogue.inactive_sell_through	true (DR-3)	Federation	cat.sku.deactivate
catalogue.assortment_core_tag	core-range	Federation	cat.tag.manage
catalogue.attributes_max_kb	4	Federation	–

8. Screens as paper

Screen	Paper mirrored	Surface	Roles	Notes
Price book (catalogue browser)	Price book	W	Steward, MPCS	Search in any script; status and tag filters; thumbnails; bulk actions on selection
Product card	Product card	W	Steward, MPCS	Three names with completeness marks; units and conversions; barcodes with batch links; images; tags; attributes; batch list tab
Stock list upload (bulk import)	Stock list	W	Steward, MPCS	Validate then commit; report with row reasons; suspects panel
Duplicate suspects	–	W	Steward	Side-by-side cards; merge or distinct; remembered decisions
Translation completeness	–	W	Steward	By tag and owner; export of missing rows for translators; import back
Batch card	Batch card	W	Warehouse, MPCS	Identity, expiry, MRP, supplier, origin document; correction opens ReasonCapture and step-up
Tax rates	Gazette	W	Federation	Categories and effective-dated rates; apply_from
Item lookup	Price list	T	Cashier	Shared ScanField and ItemCard (doc 30); read-only; sell-through badge

9. Non-functional design

Aspect	Design
9.1 Capacity	Shared catalogue: tens of thousands of SKUs; local SKUs: up to a few hundred per MPCS (hundreds of thousands federation-wide); barcodes two to three per SKU; batches: millions per year (synthetic ones dominate). Batch is partitioned by <code>created_at</code> month; lookups are by (sku, supplier, batch_no) and by id.
9.2 Snapshot size	A shop's assortment: typically 2,000–6,000 SKUs; catalogue rows about 1 KB each with barcodes and conversions; thumbnails about 8 KB each; a full snapshot under 20 MB including thumbnails, under 5 MB without (<i>doc 32 §10</i>). Deltas are per changed row.
9.3 Search latency	Trigram indexes on the three names plus sku_code; target under 100 ms for a partial name on the back office; the till searches its local copy.
9.4 Collation and normalisation	NFC at write (kernel converter); ICU collations on the name columns; sort by the display language (<i>doc 19 §5.4</i>); the transliteration table for duplicate scoring is a versioned repository asset.
9.5 Security and privacy	No personal data in the catalogue; supplier names are business names; images are public-safe product photographs; LOCAL SKU visibility is OWN plus Federation view (M-03).
9.6 Availability	Catalogue reads are on every sale path at central (ingestion re-validation) and on the till (local); the per-instance cache of SHARED SKUs by id is invalidated by sku.*.v1 events.

10. Test strategy

Layer	What is tested
Unit	Code generation uniqueness; name completeness rules per status; conversion arithmetic incl. effective dates; barcode uniqueness rules per symbology and owner; synthetic batch naming; assortment set arithmetic
Contract	Every command and query; LookupByBarcode with gtin+lot resolving a batch without a registered batch barcode; snapshot contributor output shape
State machine	SKU, barcode, batch, tax rate transitions; promotion and merge paths with alias consequences
Property-based	Random import sheets: every row either creates exactly one DRAFT or one report line; re-import is idempotent; duplicate scoring is symmetric and stable under NFC variants of the same name
Data quality	The Sinhala/Tamil name test set of doc 19 §5.4 extended with known duplicates and known distinct pairs; threshold tuned on it
Scenario	Flows 6.1–6.8; doc 13 scenario 3 (repack) and 4 (batch MRPs) for M2's part; a GRN line for a has_printed_mrp SKU without MRP is blocked
Performance	Import of 20,000 rows with validation under 5 minutes; search under 100 ms; snapshot build for a 6,000-SKU assortment under 5 s

11. Decision requests

#	Point	Proposed	Decider
DR-1	Minimal supplier table in M2 until M10 (CR-22-1)	Adopt	Architecture review
DR-2	Assortment table and core-range tag (CR-22-2)	Adopt	Architecture review; Federation names the core-range policy owner
DR-3	Deactivation while stock remains (B-17)	INACTIVE blocks new supply and pricing; tills sell through existing lots	Federation admins
DR-4	Duplicate threshold and transliteration table	0.85; table v1 curated by the steward with the language reviewers	Data steward (to be named)
DR-5	Local image override for shared SKUs	Allowed, scoped to the entity's locations	Federation
DR-6	In-store label content	SKU code + weight or batch; never price	Federation, M3 and M6 designers
DR-7	has_printed_mrp flag on SKU	Adopt (CR to doc 18): makes a null batch MRP legitimate for unpriced goods	Architecture review

12. Traceability

Section	Identifiers
3.1–3.2	FR-SKU-010, 020, 050, 070, 080; B-01, B-02, B-06, B-08, B-09, B-13, B-14; ADR-07; doc 18 B-I1; doc 20 Appendix A
3.3	FR-SKU-060; B-11, B-12; doc 18 B-I2; multi-MRP study options B, C, E
3.4	Doc 20 M2 and Appendix A (images)
3.5–3.6	FR-SKU-090, 110; B-15; G-09; J-03
3.7	FR-SKU-030, 040; B-03, B-04; ADR-08, ADR-38; doc 18 B-I8, DR-4, DR-5
3.8–3.9	Doc 32 §5.1; doc 20 M10 seams
4	FR-SKU-100; B-10, B-17
6	T-03; R-02 to R-04; doc 13 scenarios 3 and 4
9	NFR-LOC-010/020; NFR-PRF-040; M-03

End of document. Draft for design review; the contract in section 5 freezes when M3 acknowledges it, targeted end of Stage 2 week 3. Implementation guide: 22A.